

# BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

Anonymous Author(s)

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## Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer’s advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower semantic store consolidated by target- $k$  KMeans. The design target is episode completeness under same-domain interference: Soft O2 binds session siblings by attenuated score inheritance ( $\beta s$ ) rather than hard score copy, so a retrieved question can surface its advice turn without dumping an entire session into top- $k$ . Exact replay is a surface-form reference; main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level FlatIP rebuild, Soft O2 raises Answer Hit over dense FlatIP; hard hydration, which copies sibling scores without attenuation, can raise AH further by forcing siblings into top- $k$ . Soft O2-C is evaluated as a conditional cluster-binding mechanism under Mix reconstructions with off-session gold; gated Hybrid is the composed operator when thematic co-membership helps. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests that reuse the same per-query Answer Hit outcomes as the Soft O2 tables.

*Keywords:* legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

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## 1. Introduction

2 A junior associate may ask a firm assistant: “Last month you advised  
3 this client on terminating a fixed-term contract—what remedy did we rec-  
4 ommend?” The system returns several near-identical “contract termination”

5 questions, ranks the stored user utterance highly, and never surfaces the part-  
 6 ner’s written advice. The subsequent reply may sound fluent yet be grounded  
 7 in the wrong episode—or in none. In consultation archives, questions and  
 8 answers share specialized terminology, neighbouring matters are topically ad-  
 9 jacent yet legally distinct, and the professionally useful evidence unit is the  
 10 full consultation *episode*.

11 Legal informatics has long studied statutory interpretation, case retrieval,  
 12 and advisory systems Martinez-Gil (2023); Ashley (2017), and recent legal  
 13 large language models (LLMs) further improve statutory and fact-pattern  
 14 answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical  
 15 legal information retrieval (IR), however, mainly targets *external* authority—  
 16 statutes, cases, or community answers—as documents Askari et al. (2024);  
 17 Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented  
 18 dialogue systems recover evidence from evolving conversation logs Wu et al.  
 19 (2024); Maharana et al. (2024); Packer et al. (2023), yet firm assistants still  
 20 need a reliable index of *internal consultation memory*: prior client-specific  
 21 advice buried in a growing same-domain archive under lexical interference  
 22 from neighbouring matters.

23 Under that interference, dense turn retrieval often returns a question  
 24 turn while its answer sibling remains outside top- $k$ . We term this failure  
 25 *episode incompleteness*, as distinct from a full session miss. It is worsened  
 26 when approximate indexes distort turn scores and fragment episode context  
 27 at moderate archive sizes. Parent-document hydration and session-max ag-  
 28 gregation can force siblings into the result list by hard score copy, but they  
 29 also erase rank competition among unrelated candidates. Soft O2 therefore  
 30 completes missing siblings by attenuated score inheritance rather than hard  
 31 score copy.

32 This paper presents BIMS-LEGAL, a dual-store system for Chinese legal  
 33 consultation logs. A fast episodic pathway indexes timestamped turn em-  
 34 beddings with speaker role and session identifier *sid*; a slower semantic path-  
 35 way consolidates turns with target- $k$  KMeans. We adopt a dual-pathway  
 36 layout—fast episode index; slower semantic consolidation—as an architec-  
 37 tural analogy to Complementary Learning Systems McClelland et al. (1995);  
 38 Kumaran et al. (2016), without claiming a neuroscientific equivalence. On  
 39 the episodic pathway, Soft O2 performs session soft binding: when a turn  
 40 is retrieved with score  $s$ , siblings sharing *sid* inherit  $\max(s_\nu, \beta \cdot s)$ , so an  
 41 answer turn can surface under question-hit / answer-miss while attenuated  
 42 scores keep candidate competition intact. On the semantic pathway, Soft O2-

C applies the same rule within KMeans clusters when gold evidence is not co-session with the query; gated Hybrid is intended to trigger cluster binding only on direct dense hits. Exact FlatIP indexing (O1) restores faithful turn scores when product quantization collapses Answer Hit; bulk-load consolidation (O3) makes large shared-store construction tractable without changing retrieval semantics. We evaluate Soft O2 primarily against FlatIP under episode incompleteness, treat hard hydration as an unrestricted score-copy upper bound, and examine Soft O2-C under Mix reconstructions as a conditional mechanism test.

The main contributions are as follows. First, we formulate consultation-memory retrieval under same-domain interference and identify episode incompleteness as a recurrent failure mode of dense turn retrieval. Second, we propose Soft O2 as a turn-level soft-binding operator: a gap-ratio score-competition analysis identifies a feasible high- $\beta$  band  $\rho < \beta < 1$ , and an Answer Hit sensitivity sweep over  $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$  confirms a high-AH plateau from which we freeze default  $\beta=0.98$  (Table A.4), with O1/O3 as implementation operators for shared-store evaluation. Third, we evaluate Soft O2 on CAIL multi-turn and rebuilt LegalEp corpora against FlatIP, hard hydration, and shuffled-*sid* under a shared FlatIP rebuild, with BM25-turn, RRF, and cross-encoder controls as secondary comparisons, and report Soft O2-C Mix contrasts as conditional mechanism evidence with graded nDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL and the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations, and Section 7 concludes.

## 2. Related Work

### 2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside

an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al. (2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives compound that difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question–answer episode.

## 2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL indexes episodic turns with *sid* (Soft O2) and thematic KMeans clusters (Soft O2-C) when query and gold evidence occupy different sessions but share thematic structure. Under dense same-domain interference, a question turn can occupy top- $k$  while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

## 2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits  $\beta \cdot s$  for siblings and keeps

ranking competition among candidates. Parent-document and hierarchical retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores.

*Contrast with joint-episode indexing..* An alternative way to keep episode context is to concatenate full question-answer pairs into monolithic episode documents (joint indexing). Joint indexing can prevent episode incompleteness by construction, but it changes the evidence unit from individual turns to merged documents. That coarser unit weakens fine-grained turn matching—for example, isolating a specific advice sentence—and raises index memory and update cost when new turns arrive in an online consultation log. Soft O2 instead keeps turn-level representations and candidate competition, recovering missing siblings on demand via attenuated score inheritance ( $\beta s$ ) without replacing the underlying turn index. Table 1 summarizes the session-structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

| Mechanism           | Unit             | Expansion         | Type | Rank  |
|---------------------|------------------|-------------------|------|-------|
| Parent hydration    | Session          | Insert siblings   | Hard | No    |
| Session-max         | Session          | Session max       | Hard | Same* |
| Joint episode index | Session doc      | Atomic            | N/A  | N/A   |
| Soft O2             | Turn+ <i>sid</i> | Soft $\beta s$    | Soft | Yes   |
| Soft O2-C           | Turn+cluster     | Soft $\beta_{cs}$ | Soft | Yes   |

\*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the KMeans semantic pathway.

### 141 3. Method: BIMS-LEGAL

#### 142 3.1. Architecture overview

143 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-  
144 tary episodic and semantic stores over the same consultation archive.

145 **Definition 1** (Episodic and semantic elements). *An episodic turn (episodic*  
146 *element) is  $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$ , where  $\mathbf{v}_i \in \mathbb{R}^d$  is an embedding,  $t_i$  a times-*  
147 *tamp,  $c_i$  conversational context (including speaker role),  $\phi_i$  surface features,*  
148 *and  $\text{sid}_i$  the consultation session identifier. We reserve session turns for the*  
149 *weaker notion of turns that share only  $\text{sid}$  under a grouping key, without re-*  
150 *quiring role, timestamp, or other episodic metadata to participate in scoring.*  
151 *Soft O2 binds over session siblings  $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$ , but*  
152 *the indexed objects themselves remain episodic turns. A semantic element is*  
153  *$s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$  with centroid  $\mathbf{c}_j$ , member set  $C_j$ , label  $\kappa_j$ , and formation*  
154 *statistics  $\tau_j$ . Write  $\text{Clus}(t) = \{t' : \text{cid}_{t'} = \text{cid}_t\} \setminus \{t\}$  for semantic-cluster*  
155 *siblings.*

156 **Definition 2** (Dual-store retrieval map). *Given query  $q$  with embedding*  
157  *$\mathbf{v}_q$ , base episodic retrieval returns a scored candidate set  $\mathcal{R}_0(q) = \{(t, s_t)\}$*   
158 *under Eq. (1). Soft O2 produces  $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$  (Sec-*  
159 *tion 3.3). Soft O2-C produces  $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$  (Sec-*  
160 *tion 3.5). Gated Hybrid composes session binding then cross-session cluster*  
161 *binding (Algorithm 2). The returned list is the top- $k$  of the bound set under*  
162 *the rewritten scores.*

#### 163 3.2. Base scoring and implementation operators (O1, O3)

164 For query  $q$  with embedding  $\mathbf{v}_q$ , episodic base retrieval scores a memory  
165 turn  $m$  by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

166 where  $S$  is cosine similarity on  $\ell_2$ -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

167 with  $\Delta_t$  the elapsed time between query time and the turn timestamp. Unless  
168 stated otherwise,  $\alpha=0.7$ . Eq. (1) is the ranking score used throughout the  
169 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step  
170 and do not replace Eq. (1).

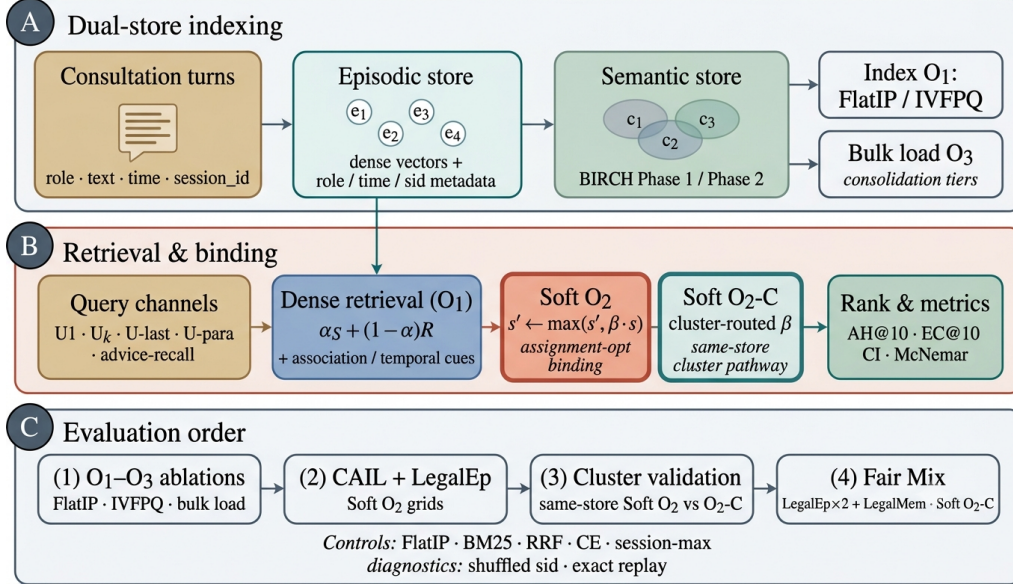


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and KMeans semantic consolidation (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

171 *O1: Exact FlatIP indexing..* Let  $\hat{S}$  denote the similarity induced by an ANN  
 172 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-  
 173 trained, so  $\|\hat{S} - S\|$  is large enough to reorder turns and break Soft O2  
 174 inheritance. O1 forces  $\hat{S} = S$  via exact FlatIP (inner-product search on  
 175  $\ell_2$ -normalized embeddings). Quantization error depresses Answer Hit (Ta-  
 176 ble A.3: IVFPQ AH 0.330 at  $M=1600$  on LegalEp-DISC exact). O1 is an  
 177 engineering prerequisite for Soft O2 evaluation, not a standalone retrieval  
 178 algorithm.

179 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-  
 180 insert update ( $e_i \mapsto \text{KMeansConsolidate}(e_i); \text{Flush}$ ) by a deferred operator  
 181  $\text{Finalize} = \text{KMeansConsolidate}(\{e_i\}_{i=1}^N) \circ \text{Flush}$ , executed once after all episodic  
 182 writes. O3 does not change retrieval semantics; it makes  $M \geq 400$  shared-  
 183 store experiments runnable ( $\approx 11$  min wall-clock for  $M=400$ ; amortized  $\approx$   
 184 0.825 s/turn over  $\sim 800$  turns). Quality-latency curves to  $M=6400$  appear  
 185 in Section 5.6.

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**Algorithm 1** Soft O2 session soft binding (implementation order)

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**Require:** Query  $q$ , dense/associative hits  $\mathcal{R}_0$ , top- $k$ , attenuation  $\beta$

- 1:  $\mathcal{R} \leftarrow \text{ExpandSiblings}(\mathcal{R}_0; \beta)$   $\triangleright$  raw-score Soft O2
  - 2:  $\mathcal{R} \leftarrow \text{ExactBoost/SparseAugment}(q, \mathcal{R})$   $\triangleright$  optional
  - 3:  $\mathcal{R} \leftarrow \text{SessionFirstRerank}(\mathcal{R}, k)$
  - 4: fuse each  $r \in \mathcal{R}$  with Eq. (1)
  - 5:  $\mathcal{R} \leftarrow \text{EnsureSessionCompleteness}(\mathcal{R}; \beta)$   $\triangleright$  final Soft O2 pass
  - 6: **return** top- $k$  of  $\mathcal{R}$  by descending fused score
- 

186 *3.3. Soft O2: session soft binding*

187 *Motivation..* Under same-domain interference, a retrieved question turn can  
188 score highly while its answer sibling remains outside top- $k$ —episode incom-  
189 pleteness compounded by IVFPQ score collapse and context fragmentation.  
190 Soft O2 completes episodes without hard score copy.

191 **Definition 3** (Soft binding operator). *Given a scored set  $\mathcal{R} = \{(t, s_t)\}$ , a*  
192 *sibling relation  $\mathcal{N}(\cdot)$ , and attenuation  $\beta \in (0, 1]$ , define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

193 *Soft O2 is  $\text{SoftBind}(\cdot; \text{Sib}, \beta)$  with default  $\beta=0.98$ . Hard parent hydration is*  
194 *the special case  $\beta=1$  with forced insertion of all siblings; session-max replaces*  
195 *each member score by  $\max_{t \in \text{sid}} s_t$ .*

196 The released Soft O2 pipeline expands session siblings on raw dense scores,  
197 optionally applies exact-match boost and sparse augment, performs session-  
198 first reranking, fuses semantic and recency scores (Eq. (1)), then runs a final  
199 completeness pass before top- $k$  truncation. Algorithm 1 records that full  
200 stage order;  $\beta$  acts on the score field present at each expansion stage in the  
201 released memory manager.

202 *3.4. Soft O2 attenuation and score competition*

203 The attenuation  $\beta$  controls how strongly a retrieved turn promotes its  
204 session siblings. Availability of the injected sibling rises with  $\beta$ , while fidelity  
205 of the direct hit falls as  $\beta \rightarrow 1$ . Table 2 summarizes the empirical gap-  
206 ratio distribution  $\rho = s_d/s_h$  on LegalEp stores under an exact-channel query  
207 proxy: median  $\rho$  lies well below one, so a high- $\beta$  point near 0.98 covers most  
208 measured distractors while remaining strictly below hard hydration ( $\beta=1$ ).



Table 2: Gap-ratio statistics on LegalEp  $M=3000$  stores (exact-channel query proxy; strongest measured non-session distractor). The feasible band  $\rho < \beta < 1$  motivates the high- $\beta$  operating region later confirmed by the AH grid search in Table A.4.

| Corpus         | $\rho_{50}$ | $\rho_{95}$ | Cover@0.90 | Default $\beta$ |
|----------------|-------------|-------------|------------|-----------------|
| LegalEp-DISC   | 0.746       | 0.854       | 0.981      | 0.98            |
| LegalEp-Lawyer | 0.662       | 0.815       | 0.998      | 0.98            |

209 A three-candidate score-competition model yields the feasible band  $\rho < \beta < 1$ ;  
 210 we then confirm the operating region with an Answer Hit sensitivity sweep  
 211 rather than treating a closed-form hinge as a selector (a zero-margin hinge  
 212 on  $\beta \in (0, 1]$  is uninformative because its rank term vanishes for all  $\beta < 1$ ).

213 **Definition 4** (Three-candidate score-competition model). *For a query whose*  
 214 *dense retrieval directly hits a gold turn  $h$  with fused score  $s_h > 0$ , let  $a$  be*  
 215 *the gold answer sibling and let  $d$  be the strongest non-session distractor in a*  
 216 *wide candidate pool, with scores  $s_a$  and  $s_d$ . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

217 *and defines the gap ratio  $\rho = s_d/s_h$ . Because  $d$  is the strongest measured*  
 218 *distractor, conditions based on  $\rho$  are conservative sufficient conditions for*  
 219 *beating that distractor, not exact top- $k$  boundary conditions for every com-*  
 220 *petitor.*

221 The binding analysis is most relevant in the soft-injection regime  $s_a \leq \beta s_h$ ,  
 222 where  $s'_a = \beta s_h$ .

223 **Proposition 1** (Feasible attenuation band). *Under Definition 4, assume*  
 224 *the soft-injection case  $s_a \leq \beta s_h$ . Then (i) availability against the measured*  
 225 *distractor requires  $\beta > \rho$ ; (ii) rank preservation of the direct hit over the*  
 226 *injected sibling requires  $\beta < 1$ , and more generally  $s'_a < s_h$ ; (iii) when  $\rho < 1$ ,*  
 227 *any  $\beta \in (\rho, 1)$  satisfies both conditions in the active injection case. If  $s_a > s_h$*   
 228 *already, rank preservation depends on the observed sibling score rather than*  
 229 *on  $\beta$  alone.*

230 *Proof.* If  $s_a \leq \beta s_h$ , then  $s'_a = \beta s_h$ . Availability follows from  $\beta s_h > s_d$ , i.e.  
 231  $\beta > \rho$ . Rank preservation requires  $s'_a < s_h$ , which reduces to  $\beta < 1$  when  
 232  $s_h > 0$ .  $\square$

233 *Soft-rank visualization and AH sensitivity sweep.* As a theoretical diagnostic  
 234 we also plot the soft pairwise surrogate

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

235 with logistic  $\sigma$ , temperature  $\tau$ , and rank weight  $\gamma$ .  $\mathcal{L}_{\text{sr}}$  visualizes the availability–  
 236 ranking trade-off under the measured  $\rho$  pool; it is not the selector. Default  
 237  $\beta=0.98$  is frozen after an Answer Hit sensitivity sweep over  $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$   
 238 on the CAIL Uk-followup and LegalEp exact/paraphrase evaluation channels  
 239 (Table A.4). The high-AH plateau  $\{0.9, 0.95, 0.98\}$  coincides with  $\rho < \beta < 1$ ;  
 240  $\beta=1$  is not better and  $\beta=0.5$  collapses binding. The sweep uses the same eval-  
 241 uation query pools as the Soft O2 grids—*not* a held-out validation split—and  
 242 comes from LegalMem-MT  $\beta$ -campaign stores. Absolute AH in Table A.4 is  
 243 therefore not comparable to the unified FlatIP rebuild (Tables 5–A.7), which  
 244 freezes  $\beta=0.98$  after selection; grid provenance (campaign ids, pool sizes,  
 245 overlap flag) ships with the manuscript package. Advice-recall inherits the  
 246 same frozen default without a separate gap-ratio fit.

247 Fair controls include hard parent hydration, session-max aggregation, and  
 248 shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max  
 249 coincide under full-sibling expansion, so tables report a single hard-expansion  
 250 baseline. Concatenating a full question–answer episode into one retrieval unit  
 251 (BM25-joint or dense joint-episode documents) changes the evidence unit  
 252 from turns to documents; we treat that design as an alternative indexing  
 253 choice rather than a Soft O2 turn-binding baseline (Section 4).

### 254 3.5. Semantic consolidation and Soft O2-C

255 Semantic consolidation uses target- $k$  KMeans over turn embeddings, pro-  
 256 ducing cluster ids consumed by Soft O2-C. Other agglomerative helpers that  
 257 may appear in the repository are unused on the Soft O2-C evaluation path.

258 **Definition 5** (KMeans semantic consolidation). *Given embeddings  $\{e_i\}$  and*  
 259 *target cluster count  $k$ , O3 runs  $(\text{labels}, \text{centers}) \leftarrow \text{KMeans}(\{e_i\}; k)$  once at*  
 260 *finalize time and assigns each turn  $t$  a cluster id  $\text{cid}_t$ , inducing  $\text{Clus}(t)$  in Def-*  
 261 *inition 1. Empty clusters are dropped and labels are remapped to a contiguous*  
 262 *range.*

263 **Definition 6** (Soft O2-C). *Soft O2-C is  $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$  with default*  
 264  *$\beta_c=0.95 < \beta$ , reflecting noisier thematic neighbourhoods than session mem-*  
 265 *bership. A per-cluster sibling budget  $B_c$  caps  $|\text{Clus}(t) \cap \text{Injected}|$  so large*  
 266 *thematic clusters cannot flood top- $k$ .*

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**Algorithm 2** Gated Hybrid Soft O2 + Soft O2-C

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**Require:** Base hits  $\mathcal{R}_0$ , budgets  $(k, B_c)$ , attenuations  $(\beta, \beta_c)$

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$   $\triangleright$  session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$   $\triangleright$  direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

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267 Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings  
268 relative to sessions already covered by dense /Soft O2 hits, and triggers  
269 cluster binding only on direct dense hits. This prevents thematic confusers  
270 from displacing same-session siblings already recovered by Soft O2. Soft O2  
271 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway  
272 operator when gold evidence lies in another session of the same thematic  
273 cluster.

## 274 4. Corpora and Evaluation Protocol

275 Figure 2 sketches the shared-corpus design used throughout the legal eval-  
276 uation. Gold needles and same-domain distractors occupy one store. Each  
277 query probes that shared archive under fixed interference conditions. Pri-  
278 vately rebuilt haystacks are avoided so that system comparisons isolate the  
279 retrieval operator.

### 280 4.1. Datasets and query channels

281 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.  
282 Gold needles are authentic preliminary and final dialogues, with conversation  
283 turns completed by label turns where applicable. Query channels cover the  
284 first user turn (U1), a later in-dialogue user turn (Uk-followup), the last user  
285 turn (U-last), and constrained paraphrases (U-pa). Evidence defaults to  
286 assistant turns in the target session.

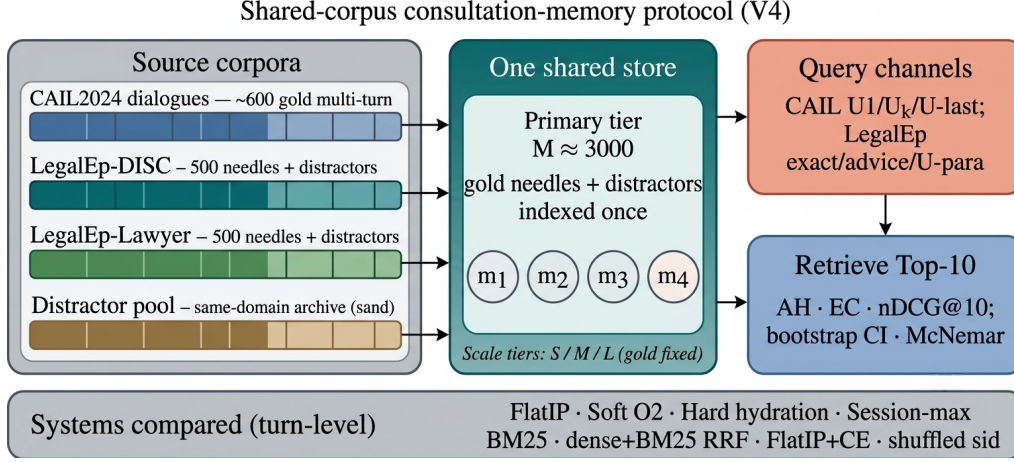


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is a natural (question, answer) pair with session identifier *sid*. Reconstruction applies length bounds and legal-content filters, excludes exam prompts, removes near-duplicates, separates needles from distractors under a fixed seed, and retains an audit subsample. After filtering, each LegalEp corpus retains 3500 passed sessions (500 needles, 2500 distractors at the  $M=3000$  primary tier).

*Query-channel policy.* Exact replay uses the stored user question and is reported only as a *diagnostic* upper bound on surface-form recovery. Primary claims use paraphrase, Uk-followup /U-last, and advice-recall queries whose surface form is not identical to the stored question text; paraphrase and advice-recall templates are generated under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

Archive size is varied with gold needles held fixed. Unless stated otherwise, Soft O2 transfer tables use the  $M \approx 3000$  tier; O1–O2 ablations use  $M=400$ ,  $S=300$ .

## 305 4.2. Metrics and baselines

306 Primary metrics are Answer Hit@ $k$  (AH), Episode Completeness@ $k$  (EC),  
 307 and normalized Discounted Cumulative Gain@ $k$  (nDCG) at  $k=10$ . AH is  
 308 binary: whether a gold answer turn appears in the top- $k$ . EC is the mean  
 309 coverage of gold turns within the target session. Session Hit@ $k$  is reported  
 310 only for diagnosis. For nDCG we use fixed graded relevance over the full  
 311 gold session: answer turns  $rel=1.0$ , other same-session turns  $rel=0.5$ , all  
 312 other turns  $rel=0.0$ . The ideal denominator (IDCG) is computed from all  
 313 gold-relevant turns in the target session and truncated at  $k$ , so every system  
 314 on a query shares the same IDCG. Answer Hit and Episode Completeness  
 315 are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session  
 316 IDCG so that systems remain comparable on each query. Hard expansion  
 317 can raise nDCG when copied siblings occupy early ranks, so AH and EC are  
 318 read together with nDCG.

319 The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL and  
 320 LegalEp transfer channels. AH, EC, nDCG, failure rates, and Table A.2 (Mc-  
 321 Nemar mid- $p$  with Holm adjustment) share one FlatIP rebuild and the same  
 322 per-query Answer Hit outcomes; each  $\Delta$ AH in Table A.2 equals Soft O2–FlatIP  
 323 in the Soft O2 tables up to ordinary rounding. Appendix Soft O2 grids re-  
 324 port AH/EC/nDCG point estimates from that rebuild without separate CI  
 325 columns. Hard hydration, CE, and Soft O2-C Mix contrasts are secondary.

326 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies  
 327 soft sibling score inheritance on the episodic pathway; Soft O2-C applies the  
 328 same rule within KMeans clusters. Hard parent hydration expands a hit  
 329 to all siblings by copying the trigger score. Lexical controls include BM25  
 330 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same  
 331 turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve  
 332 pre-concatenated episode documents and therefore answer a coarser retrieval  
 333 problem; they are discussed as alternative designs in Section 6.3 rather than  
 334 Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the  
 335 top-30 FlatIP candidates with `BAAI/bge-reranker-v2-m3`. Shuffled *sid*, a  
 336  $\beta$  sweep, and IVFPQ provide membership, attenuation, and quantization  
 337 controls.

## 338 4.3. Fair Mix protocol for Soft O2-C

339 On fully same-session LegalMem grids Soft O2 already recovers gold an-  
 340 swers that share *sid*, leaving Soft O2-C little exclusive headroom and expos-  
 341 ing it to thematic confusers. To study off-session recovery we rebuild the

342 *same* LegalEp and LegalMem archives under a Mix protocol: 70% of gold  
 343 episodes are split into query-side  $S_a$  and evidence-side  $S_b$  with distinct ses-  
 344 sion identifiers, while 30% remain intact. Distractors are unchanged. FlatIP,  
 345 Soft O2, Soft O2-C, and gated Hybrid share unrestricted dense retrieval on  
 346 this store; we report overall AH and stratum AH on cross-session vs same-  
 347 session queries.

348 *Cluster co-membership and purity under Mix..* After KMeans consolidation,  
 349 each cross-session Sa/Sb pair is assigned a shared cluster identifier. This  
 350 construction uses split metadata available only when building the evaluation  
 351 store; it does not inject labels at query time. The intent is to test Soft O2-  
 352 C when thematic co-membership is present. Cluster purity still matters: a  
 353 shared *cid* that also contains many thematically adjacent distractors narrows  
 354 Soft O2-C’s effective margin, which is why Hybrid gates cluster injection to  
 355 direct dense hits and caps the per-cluster budget  $B_c$ . If recovery still fails  
 356 under guaranteed Sa/Sb co-membership, the failure is attributable to binding  
 357 rather than accidental cluster separation. Natural same-cluster rates without  
 358 this construction are lower (Section 6.3). Soft O2’s session membership rule  
 359 is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses  
 360 mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior  
 361 runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon  
 362 memory context only Wu et al. (2024); Maharana et al. (2024).

## 363 5. Experiments

364 Unless noted, all dense retrieval grids use the multilingual encoder **nomic-**  
 365 **embed-text-v2-moe** Nussbaum et al. (2025) (768-d) as a single shared em-  
 366 bedding backend, with  $k=10$  and seed 42 on the main  $M \approx 3000$  stores. The  
 367 encoder is held fixed so Soft O2 vs FlatIP contrasts isolate binding rather  
 368 than representation changes; sensitivity to alternative Chinese or multilingual  
 369 encoders (e.g., **bge-large-zh-v1.5**) is discussed in Section 6.3. A five-seed  
 370 check on a smaller shared store ( $M=400$ ) is reported later. Primary contrasts  
 371 appear in Figures 3–5; full numeric grids are in the appendix. Each main  
 372 table maps to a released JSON artifact in the accompanying repository.

### 373 5.1. Implementation operators and Soft O2 ablation

374 Table 3 reports the shared-store ablation on the DISC-Law and Lawyer-  
 375 LLaMA corpora ( $M=400$ ,  $S=300$ ). IVFPQ baseline Answer Hit is 0.540 on

DISC and 0.397 on Lawyer. O1 alone yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397). Exact indexing is therefore a necessary prerequisite, but it is insufficient alone. O1+Soft O2 raises Answer Hit to 0.780 (DISC) and 0.680 (Lawyer), gains of +0.240 and +0.283 over baseline, with Episode Completeness 0.888 and 0.840. O3 is enabled for all configurations in this ladder as a construction prerequisite; it does not alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store ( $M=400$ ,  $S=300$ ,  $k=10$ ). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the  $M \approx 3000$  grids appears in Table 6.

| Corpus       | System          | EC@10        | AH@10        |
|--------------|-----------------|--------------|--------------|
| DISC-Law     | IVFPQ baseline  | 0.770        | 0.540        |
|              | + O1 FlatIP     | 0.782        | 0.567        |
|              | + O1+O2 Soft O2 | <b>0.888</b> | <b>0.780</b> |
| Lawyer-LLaMA | IVFPQ baseline  | 0.698        | 0.397        |
|              | + O1 FlatIP     | 0.698        | 0.397        |
|              | + O1+O2 Soft O2 | <b>0.840</b> | <b>0.680</b> |

### 5.2. Failure taxonomy

We classify each query into four mutually exclusive modes: *complete* (question and answer in top- $k$ ), *incomplete* (question hit / answer miss), *answer-only* (answer hit / question miss), and *session miss* (no gold-session turn in top- $k$ ). Table 4 reports these rates for FlatIP, Soft O2, hard hydration, and shuffled *sid* on primary non-exact channels under the unified FlatIP rebuild. Soft O2 converts many incomplete FlatIP cases into complete retrieval while session-miss stays nearly fixed; shuffled *sid* restores high incompleteness. Hard hydration can drive incompleteness near zero by score copy and often leads Soft O2 on complete rate (Section 5.4).

### 5.3. Turn-level Soft O2 controls

Primary Soft O2 claims use a shared turn-level index. Table 5 reports Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel; shuffled *sid* collapses toward or below FlatIP. Hard hydration—full sibling score copy—can exceed Soft O2 AH on the same rebuild (Table 5);

Table 4: Failure taxonomy at  $k=10$  (values in %; rows sum to 100 within rounding). DISC/Lawyer denote LegalEp corpora. Classes: complete (Q+A), incomplete (Q hit / A miss), answer-only, and session miss. Boldface: best complete / lowest incomplete among FlatIP, Soft O2, and shuffled *sid*; hard hydration is unrestricted score copy.

| Channel         | System   | Comp.       | Inc.       | A-only | Miss |
|-----------------|----------|-------------|------------|--------|------|
| DISC / u-para   | FlatIP   | 68.8        | 28.8       | 0.8    | 1.6  |
| DISC / u-para   | Soft O2  | <b>88.1</b> | <b>9.9</b> | 0.2    | 1.8  |
| DISC / u-para   | Hard     | 97.2        | 0.0        | 0.0    | 2.8  |
| DISC / u-para   | Shuf. O2 | 61.6        | 35.8       | 0.4    | 2.2  |
| DISC / advice   | FlatIP   | 47.2        | 17.6       | 0.0    | 35.2 |
| DISC / advice   | Soft O2  | <b>61.0</b> | <b>3.6</b> | 0.0    | 35.4 |
| DISC / advice   | Hard     | 64.8        | 0.0        | 0.0    | 35.2 |
| DISC / advice   | Shuf. O2 | 43.4        | 21.2       | 0.0    | 35.4 |
| Lawyer / u-para | FlatIP   | 66.9        | 29.5       | 2.8    | 0.8  |
| Lawyer / u-para | Soft O2  | <b>94.6</b> | <b>2.8</b> | 1.0    | 1.6  |
| Lawyer / u-para | Hard     | 97.8        | 0.0        | 0.0    | 2.2  |
| Lawyer / u-para | Shuf. O2 | 64.9        | 30.7       | 2.6    | 1.8  |
| CAIL / Uk       | FlatIP   | 43.2        | 56.7       | 0.0    | 0.2  |
| CAIL / Uk       | Soft O2  | <b>93.2</b> | <b>6.7</b> | 0.0    | 0.2  |
| CAIL / Uk       | Hard     | 99.7        | 0.2        | 0.0    | 0.2  |
| CAIL / Uk       | Shuf. O2 | 31.3        | 68.5       | 0.0    | 0.2  |

399 Soft O2 injects siblings at attenuated scores  $\beta s$  with  $\beta < 1$  (Tables A.5–A.7).  
400 BM25-turn is a same-granularity lexical control (Table A.8). Joint episode /  
401 document-level indexes change evidence granularity and are scoped as alter-  
402 native designs rather than Soft O2 turn-binding baselines (Section 6.3).

#### 403 5.4. Soft O2 on session (CAIL and LegalEp)

404 Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn  
405 channels at  $M \approx 3000$  under O1 FlatIP. Soft O2 raises Answer Hit over  
406 FlatIP on every channel (U1 0.948 vs 0.803; Uk 0.932 vs 0.432; U-last 0.957  
407 vs 0.393). Shuffled *sid* removes the gain, tying the improvement to correct  
408 episode membership. Hard hydration can raise AH further by copying sibling  
409 scores without attenuation (e.g., Uk 0.997 vs Soft O2 0.932). Table 6 pre-  
410 serves the Soft O2>FlatIP ordering under graded nDCG with gold-session  
411 IDCG, and incompleteness falls sharply (Table 4). Full numeric grids appear  
412 in Tables A.5–A.7.



Table 5: Primary Soft O2 controls (AH@10,  $M \approx 3000$ , unified FlatIP rebuild). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is compared primarily to FlatIP; hard hydration is unrestricted score copy; shuffled *sid* tests membership dependence.

| Corpus / channel        | FlatIP | Soft O2 | Hard         | Shuffled |
|-------------------------|--------|---------|--------------|----------|
| CAIL / U1               | 0.803  | 0.948   | <b>0.995</b> | 0.660    |
| CAIL / Uk               | 0.432  | 0.932   | <b>0.997</b> | 0.313    |
| CAIL / U-last           | 0.393  | 0.957   | <b>0.998</b> | 0.255    |
| LegalEp-DISC / u-para   | 0.696  | 0.883   | <b>0.972</b> | 0.620    |
| LegalEp-DISC / advice   | 0.472  | 0.610   | <b>0.648</b> | 0.434    |
| LegalEp-Lawyer / u-para | 0.697  | 0.956   | <b>0.978</b> | 0.675    |
| LegalEp-Lawyer / advice | 0.416  | 0.654   | <b>0.674</b> | 0.396    |

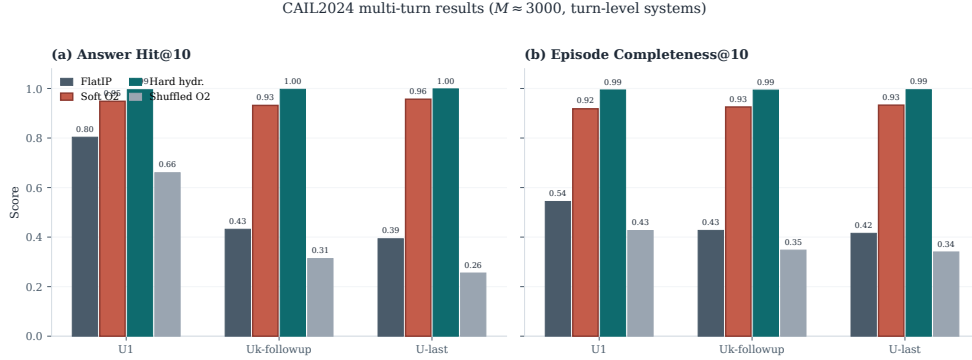


Figure 3: CAIL2024 multi-turn results at tier  $M \approx 3000$ . Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard hydration can further raise AH/EC.

Figure 4 reports LegalEp transfer on the same AH grids. Soft O2 also raises exact replay (DISC 0.916 vs FlatIP 0.714; Lawyer 0.976 vs 0.678), which we treat as diagnostic relative to paraphrase and advice-recall. Paraphrase gains are large: 0.883 vs 0.696 on DISC and 0.956 vs 0.697 on Lawyer. Advice-recall rises to 0.610 from FlatIP 0.472 on DISC and to 0.654 from 0.416 on Lawyer. Hard hydration can still edge Soft O2 on these channels; shuffled *sid* collapses toward FlatIP. Graded nDCG mirrors the Soft O2>FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed grids are in Tables A.6–A.7.

Cross-encoder reranking is a strong neural control at the same turn-level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with bge-

Table 6: Graded nDCG@10 with gold-session IDCG (answer  $rel=1.0$ , other same-session turns  $rel=0.5$ ) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

| Corpus         | Channel | System      | nDCG@10      | Incomplete |
|----------------|---------|-------------|--------------|------------|
| CAIL           | Uk      | FlatIP      | 0.428        | 56.7%      |
| CAIL           | Uk      | Soft O2     | <b>0.796</b> | 6.7%       |
| CAIL           | Uk      | Shuffled O2 | 0.349        | 68.5%      |
| CAIL           | U-last  | FlatIP      | 0.400        | 60.5%      |
| CAIL           | U-last  | Soft O2     | <b>0.796</b> | 4.2%       |
| CAIL           | U-last  | Shuffled O2 | 0.326        | 74.3%      |
| LegalEp-DISC   | u-para  | FlatIP      | 0.659        | 28.8%      |
| LegalEp-DISC   | u-para  | Soft O2     | <b>0.742</b> | 9.9%       |
| LegalEp-DISC   | u-para  | Shuffled O2 | 0.590        | 35.8%      |
| LegalEp-DISC   | advice  | FlatIP      | 0.454        | 17.6%      |
| LegalEp-DISC   | advice  | Soft O2     | <b>0.518</b> | 3.6%       |
| LegalEp-DISC   | advice  | Shuffled O2 | 0.411        | 21.2%      |
| LegalEp-Lawyer | u-para  | FlatIP      | 0.708        | 29.5%      |
| LegalEp-Lawyer | u-para  | Soft O2     | <b>0.832</b> | 2.8%       |
| LegalEp-Lawyer | u-para  | Shuffled O2 | 0.656        | 30.7%      |
| LegalEp-Lawyer | advice  | FlatIP      | 0.457        | 26.2%      |
| LegalEp-Lawyer | advice  | Soft O2     | <b>0.569</b> | 2.0%       |
| LegalEp-Lawyer | advice  | Shuffled O2 | 0.424        | 27.8%      |

reranker-v2-m3. Table 7 reports a secondary CE campaign (separate from the unified Soft O2 rebuild): within that campaign FlatIP+CE is strongest on LegalEp exact channels, while Soft O2 leads on CAIL multi-turn channels (Uk 0.698 vs CE 0.375; U-last 0.762 vs 0.342). A structural limit of the reranker explains the split: CE can refine ranking among candidates already retrieved by FlatIP, but it cannot promote an answer sibling that never entered the top-30 pool. Soft O2 recovers such siblings by propagating attenuated scores through session links, including cases where the stored advice turn has weak direct lexical overlap with the query (paraphrase and advice-recall; Tables 5 and 6).

#### 5.5. Soft O2-C and gated Hybrid under Mix

When gold answers already share *sid* with the query, Soft O2 is sufficient and cluster binding is unnecessary. Table 8 asks the off-session question under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated Hybrid sharing

LegalEp ( $M=3000$ ): Soft O2 gains are strongest on paraphrase and remain positive on advice-recall

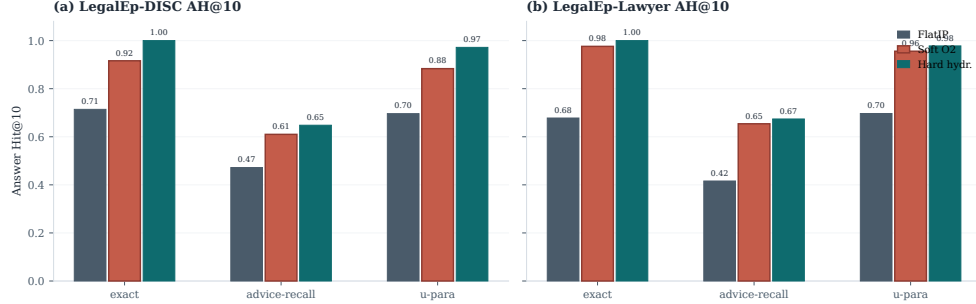


Figure 4: LegalEp AH@10 ( $M=3000$ , 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Secondary cross-encoder control (AH@10; CE campaign store). FlatIP+CE reranks the top-30 FlatIP candidates with a multilingual cross-encoder. Soft O2 column is from the same CE campaign for within-table comparison; primary Soft O2 vs FlatIP AH uses the unified rebuild (Tables 5–A.7). Boldface: best AH within each row.

| Corpus / channel       | FlatIP | FlatIP+CE    | Soft O2      |
|------------------------|--------|--------------|--------------|
| CAIL / U1              | 0.585  | 0.688        | <b>0.788</b> |
| CAIL / Uk              | 0.265  | 0.375        | <b>0.698</b> |
| CAIL / U-last          | 0.228  | 0.342        | <b>0.762</b> |
| LegalEp-DISC / exact   | 0.572  | <b>0.722</b> | 0.572        |
| LegalEp-Lawyer / exact | 0.636  | <b>0.708</b> | 0.636        |

unrestricted dense retrieval. Hybrid uses a direct-dense gate: cluster binding fires only on turn ids recorded *before* Soft O2 session expansion.

On LegalEp Mix stores, gated Hybrid is the strongest overall system (DISC AH 0.542, Lawyer 0.536; McNemar mid- $p=0.0005^{**}$  /  $0.0022^{**}$  vs Soft O2) and also leads  $AH_{\text{cross}}$ . Ungated Soft O2-C raises DISC  $AH_{\text{cross}}$  to 0.506 from Soft O2 0.460 and Lawyer cross to 0.480 from 0.440, but remains weaker than Hybrid once session binding is composed under the gate. On LegalMem-MT Mix the picture reverses: Soft O2 retains the best overall AH (0.602), while Soft O2-C (0.440) and Hybrid (0.562) trail; FlatIP even leads the cross stratum (0.520). Cluster binding is therefore conditional on corpus and cluster purity—helpful as a gated add-on on LegalEp advice-recall Mix, but not uniformly beneficial on CAIL multi-turn Mix under the same protocol. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led

452 (Appendix Appendix A, Table A.1).

Table 8: Mix evaluation of Soft O2-C and gated Hybrid with the direct-dense cluster trigger (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: \*  $p < 0.05$ , \*\*  $p < 0.01$ .

| Corpus             | System    | AH           | AH <sub>cross</sub> | AH <sub>same</sub> | mid- $p$ vs Soft O2 |
|--------------------|-----------|--------------|---------------------|--------------------|---------------------|
| LegalEp-DISC Mix   | FlatIP    | 0.494        | 0.491               | 0.500              | —                   |
|                    | Soft O2   | 0.492        | 0.460               | 0.567              | —                   |
|                    | Soft O2-C | 0.498        | 0.506               | 0.480              | 0.7275              |
|                    | Hybrid    | <b>0.542</b> | <b>0.526</b>        | <b>0.580</b>       | 0.0005**            |
| LegalEp-Lawyer Mix | FlatIP    | 0.460        | 0.451               | 0.480              | —                   |
|                    | Soft O2   | 0.492        | 0.440               | <b>0.613</b>       | —                   |
|                    | Soft O2-C | 0.464        | 0.480               | 0.427              | 0.1284              |
|                    | Hybrid    | <b>0.536</b> | <b>0.520</b>        | 0.573              | 0.0022**            |
| LegalMem-MT Mix    | FlatIP    | 0.530        | <b>0.520</b>        | 0.553              | —                   |
|                    | Soft O2   | <b>0.602</b> | 0.489               | <b>0.867</b>       | —                   |
|                    | Soft O2-C | 0.440        | 0.460               | 0.393              | 0.0000**            |
|                    | Hybrid    | 0.562        | 0.446               | 0.833              | 0.0594              |

453 Hybrid uses the direct-dense gate. Soft O2-C mid- $p$  vs Soft O2 may favor either side; on  
 454 LegalMem-MT the \*\* marks Soft O2-C significantly *below* Soft O2. Cluster co-membership  
 455 of Sa/Sb pairs follows Section 4.3.

### 456 5.6. Scale behaviour

457 Figure 5 and Table A.3 report quality and warm-query latency on the  
 458 LegalEp DISC exact channel ( $Q=100$ ; three repeats; archive sizes 100, 400,  
 459 1600, and 6400). Soft O2 retains an AH advantage at each tier. IVFPQ  
 460 collapses at  $M=1600$  (AH 0.330), which motivates O1, while absolute latency  
 461 grows with  $M$ . We treat O1 FlatIP as an engineering prerequisite for Soft O2  
 462 at these scales and do not claim deployment readiness beyond the measured  
 463 curve.

464 *End-to-end QA audit (summary)*.. A dual-protocol QA audit ( $N=270$  per  
 465 corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) reports judged  
 466 answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs  
 467 Soft O2 generation experiment. Full protocol and stratified results appear in  
 468 Section 6.2. On a smaller shared store ( $M=400$ , five seeds), Soft O2 AH on  
 469 LegalEp-DISC is  $0.851 \pm 0.008$  versus FlatIP  $0.698 \pm 0.040$ .

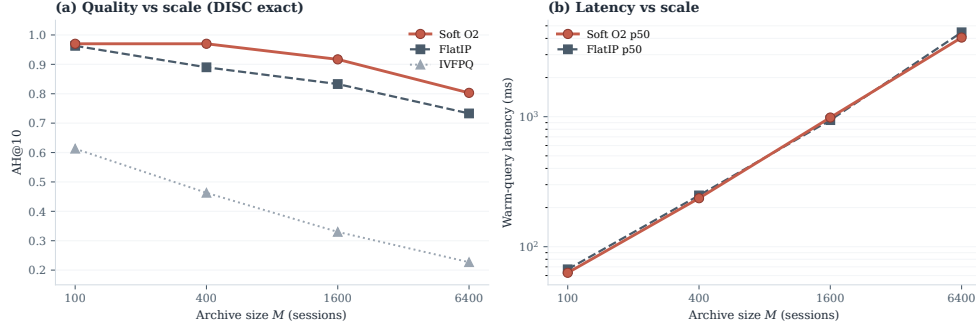


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as  $M$  grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate  $M$ .

## 6. Discussion

### 6.1. Findings and scope

Episode incompleteness is a substantial FlatIP failure under same-domain interference; Soft O2 converts many incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through soft sibling inheritance; O3 makes large shared stores tractable. The gap-ratio band  $\rho < \beta < 1$  and the AH sensitivity plateau  $\{0.9, 0.95, 0.98\}$  jointly motivate freezing  $\beta = 0.98$  below hard hydration. Primary Soft O2 vs FlatIP gains on CAIL and LegalEp remain Holm-significant (Table A.2); paraphrase and advice-recall are the main transfer channels, with graded nDCG agreeing (Tables 5–6). Under Mix, gated Hybrid improves overall AH over Soft O2 on LegalEp-DISC and LegalEp-Lawyer, while Soft O2 remains strongest on LegalMem-MT and on intact same-session gold (Tables 8–A.1). Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes remain a coarser design outside this protocol.

### 6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool  $N=270$  judged instances as P1  $n=120$  plus P2  $n=150$ . The

independent judge assigns a continuous evidence-answerability score in  $[0, 1]$  from the retrieved context and the generated reply; Evidence-Answerability Rate (EAR) is the fraction of instances with judge score  $\geq 0.7$ . Table 9 reports pooled EAR and the mean judge score on a stratified subsample ( $n=90$ /corpus; 30 each of exact, paraphrase, and follow-up), labelled “Stratified judge” in the table. In the complete-episode stratum—both question and answer turns in top- $k$ —judged answerability is higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI  $[0.83, 0.90]$  /  $[0.82, 0.89]$ ). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ( $n=60$ /corpus) separates retrieval failures (gold episode absent from top- $k$ ) from generation failures given retrieved evidence. Annotators used a binary answerability label on each instance; agreement is Cohen’s  $\kappa$  computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero: grounding in retrieved episodes does not guarantee statutory correctness. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit ( $N=270$  per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score  $\geq 0.7$ . Stratified judge: mean judge score on  $n=90$ /corpus (exact/paraphrase/follow-up). Human  $\kappa$ : Cohen’s  $\kappa$  on binary answerability labels from two blind annotators ( $n=60$ /corpus).

| Corpus       | Pooled EAR   | Stratified judge | Human $\kappa$ |
|--------------|--------------|------------------|----------------|
| DISC-Law     | <b>0.867</b> | 0.757            | 0.71           |
| Lawyer-LLaMA | <b>0.859</b> | 0.729            | 0.68           |

### 6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL.

520 *Dependency on session segmentation and cluster quality..* Soft O2 relies on  
521 correct session identifiers (*sid*). In uncured client logs, boundary noise—  
522 mid-session topic shifts, multi-intent interruptions, or automated segmenta-  
523 tion drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated  
524 inheritance may propagate false context; if a true advice turn is split into an-  
525 other session, Soft O2 cannot bind it. Soft O2-C likewise depends on KMeans  
526 cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared cluster  
527 id so binding can be tested when thematic co-membership is present (Sec-  
528 tion 4.3); in fully unsupervised consolidation, noisier boundaries shrink the  
529 margin over dense retrieval. Deployments on raw consultation streams there-  
530 fore need reliable upstream session segmentation, and Soft O2-C should be  
531 treated as gated rather than always-on.

532 *Joint-episode indexing..* Concatenating full question-answer pairs into  
533 episode documents prevents incompleteness by changing the evidence unit  
534 from turns to merged documents (Section 2). That design lies outside the  
535 Soft O2 turn-binding protocol. A matched comparison under online updates,  
536 latency, and fine-grained advice isolation is left to future work.

537 *Embedding model and other scope limits..* All main grids use **nomic-embed-**  
538 **text-v2-moe**. A Chinese-specialized encoder such as **bge-large-zh-v1.5**  
539 could alter absolute dense ranks and the measured  $\rho$  distribution, and might  
540 shrink or widen Soft O2’s relative gain wherever the bi-encoder already places  
541 answer turns inside top- $k$ . We did not re-run the full Soft O2 grids under al-  
542 ternative encoders; the binding operator itself is encoder-agnostic once fused  
543 scores are available. The  $\rho$  statistics use an exact-channel query proxy and  
544 the strongest measured distractor; paraphrase and advice-recall gap laws may  
545 shift the feasible band slightly, though the AH sweep keeps a stable high- $\beta$   
546 region on those channels. Main  $M \approx 3000$  tables use a fixed seed; five-seed vari-  
547 ance is reported only for the smaller  $M=400$  store. LLM paraphrases need  
548 human audit. Statutory correctness, citation validity, and professional liabil-  
549 ity remain unevaluated. Because many channels are tested, Holm-adjusted  
550  $p$ -values accompany the Soft O2 vs FlatIP family (Table A.2); Mix Hybrid sig-  
551 nificance is reported only vs Soft O2 within Table 8 and is corpus-conditional.

552 *Latency and deployment scope..* Scale curves cover archive sizes 100, 400,  
553 1600, and 6400 under a fixed hardware stack (Table A.3). At  $M=6400$ , warm-  
554 query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms for FlatIP)—  
555 unacceptable for millisecond interactive search. BIMS-LEGAL is there-

fore best positioned for high-precision offline or near-real-time consultation-memory recovery—for example, pre-meeting case review, audit replay, or batch RAG over a firm’s advice archive—rather than as a sub-millisecond search engine. Future work will combine approximate nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top of faithful candidate retrieval, preserving episode completion while reducing query latency. Behavior at roughly  $10^5$  sessions is not measured, and we do not claim asymptotic scalability from O3 wall-clock alone.

## 7. Conclusion

BIMS-LEGAL recovers prior advice from Chinese legal consultation archives under same-domain interference with a dual store: an episodic turn index and a KMeans semantic store. Soft O2 completes missing siblings by attenuated session binding ( $\beta=0.98$ ), selected inside the gap-ratio band via an Answer Hit sensitivity sweep (Table A.4); hard hydration remains a stronger unattenuated upper bound. Soft O2-C extends the same operator to thematic clusters under Mix reconstructions as a conditional mechanism test, with O1 FlatIP and O3 bulk-load as the shared-store implementation operators. On CAIL multi-turn and LegalEp paraphrase /advice-recall channels, Soft O2 improves Answer Hit and graded nDCG over FlatIP and cuts incompleteness; shuffled-*sid* collapses the Soft O2 gain. Under Mix, gated Hybrid recovers additional LegalEp gold when cluster co-membership is present, while Soft O2 remains preferable on LegalMem-MT Mix and on intact same-session gold. A Soft O2 QA audit links complete-episode retrieval to judged answerability with low wrong-session contamination; it is not a paired FlatIP vs Soft O2 generation comparison.

## Declarations

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Computational resources were provided by the authors’ host institutions.

### *Conflict of interest*

The authors declare no competing interests.



586 *Ethics approval*

587 This study uses publicly released research corpora and excludes confidential client data. The system is a research prototype for consultation-memory  
588 retrieval.  
589

590 *Data availability*

591 Code, processed evaluation corpora, and primary Soft O2 metric artifacts  
592 (including paired per-query Answer Hit used for Holm tests) are available at  
593 URL withheld for anonymous review. (archived release to accompany camera-  
594 ready). Upstream corpora remain under their original licenses: CAIL2024  
595 consultation tracks; Hugging Face ShengbinYue/DISC-Law-SFT and Skepsu  
596 n/lawyer\_llama\_data. Processed LegalEp /LegalMem artifacts are derived  
597 research releases; redistributors should consult each upstream licence before  
598 reuse.

599 *Code availability*

600 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load,  
601 Soft O2-C), corpus builders, Mix constructors, and the scripts that regenerate  
602 Soft O2 tables, figures, and Holm tests from one metric family ship in the  
603 same repository.

604 *Author contributions*

605 Author contributions are withheld for anonymous review.

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720 **Appendix A. Numeric grids and controls**

721 Same-session Soft O2-C contrasts, significance tests, scale curves, and full  
 722 Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

| Corpus / channel        | Soft O2      | Soft O2-C |
|-------------------------|--------------|-----------|
| LegalEp-DISC / exact    | <b>0.836</b> | 0.744     |
| LegalEp-DISC / u-para   | <b>0.807</b> | 0.730     |
| LegalEp-DISC / advice   | <b>0.548</b> | 0.490     |
| LegalEp-Lawyer / exact  | <b>0.818</b> | 0.710     |
| LegalEp-Lawyer / u-para | <b>0.817</b> | 0.733     |
| LegalEp-Lawyer / advice | <b>0.546</b> | 0.448     |
| LegalMem-MT / U1        | <b>0.903</b> | 0.712     |
| LegalMem-MT / Uk        | <b>0.793</b> | 0.385     |

Table A.2: Primary Soft O2 vs FlatIP family: McNemar mid- $p$  on paired Answer Hit from the unified FlatIP rebuild, with Holm step-down adjustment ( $m=9$ ). Each  $\Delta\text{AH}$  matches Soft O2–FlatIP in the main Soft O2 tables; paired outcomes are released with the paper artifacts.

| Corpus / channel        | Contrast     | $\Delta\text{AH}$ | mid- $p$ | Holm $p$ |
|-------------------------|--------------|-------------------|----------|----------|
| CAIL / Uk               | O2 vs FlatIP | +0.500            | 3.69e-69 | 2.95e-68 |
| CAIL / U1               | O2 vs FlatIP | +0.145            | 1.15e-16 | 3.44e-16 |
| CAIL / U-last           | O2 vs FlatIP | +0.563            | 3.19e-85 | 2.87e-84 |
| LegalEp-DISC / exact    | O2 vs FlatIP | +0.202            | 2.32e-19 | 9.28e-19 |
| LegalEp-Lawyer / exact  | O2 vs FlatIP | +0.298            | 1.16e-39 | 8.12e-39 |
| LegalEp-DISC / u-para   | O2 vs FlatIP | +0.187            | 1.28e-16 | 3.44e-16 |
| LegalEp-Lawyer / u-para | O2 vs FlatIP | +0.259            | 2.07e-32 | 1.04e-31 |
| LegalEp-DISC / advice   | O2 vs FlatIP | +0.138            | 1.52e-15 | 1.52e-15 |
| LegalEp-Lawyer / advice | O2 vs FlatIP | +0.238            | 7.84e-33 | 4.70e-32 |

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each  $M$  block; lowest p50/p95 within each  $M$  block.

| $M$  | Mode    | AH@10        | Build (s)     | p50 (ms)    | p95 (ms)    |
|------|---------|--------------|---------------|-------------|-------------|
| 100  | FlatIP  | 0.963        | 3.0           | 67          | 80          |
| 100  | IVFPQ   | 0.613        | <b>25.2</b>   | <b>40</b>   | <b>54</b>   |
| 100  | Soft O2 | <b>0.970</b> | 3.2           | 63          | 69          |
| 400  | FlatIP  | 0.890        | 26.5          | 248         | 305         |
| 400  | IVFPQ   | 0.463        | 89.5          | <b>171</b>  | <b>204</b>  |
| 400  | Soft O2 | <b>0.970</b> | 30.9          | 236         | 313         |
| 1600 | FlatIP  | 0.833        | 359.8         | <b>941</b>  | 1278        |
| 1600 | IVFPQ   | 0.330        | <b>251.4</b>  | 669         | <b>751</b>  |
| 1600 | Soft O2 | <b>0.917</b> | 393.7         | 985         | 1172        |
| 6400 | FlatIP  | 0.733        | <b>4722.8</b> | 4458        | 4969        |
| 6400 | IVFPQ   | 0.227        | 1072.1        | <b>2182</b> | <b>3021</b> |
| 6400 | Soft O2 | <b>0.803</b> | 5245.7        | 4061        | 4503        |

Table A.4: Soft O2 AH@10 sensitivity sweep over  $\beta \in \{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$  on the CAIL Uk-followup and LegalEp exact/paraphrase evaluation channels (LegalMem-MT  $\beta$ -campaign stores; not held-out). Used with the gap-ratio analysis to freeze default  $\beta=0.98$ . Peak AH lies in  $\{0.9, 0.95, 0.98\}$ ;  $\beta=0.5$  weakens binding. Not a primary Soft O2 result table; absolute AH values are not comparable to the unified FlatIP-rebuild Tables 5–A.7.

| Corpus                  | 0.5   | 0.7   | 0.9   | 0.95  | 0.98  | 1.0   |
|-------------------------|-------|-------|-------|-------|-------|-------|
| CAIL / Uk               | 0.323 | 0.537 | 0.697 | 0.700 | 0.698 | 0.662 |
| LegalEp-Lawyer / exact  | 0.608 | 0.654 | 0.754 | 0.754 | 0.752 | 0.752 |
| LegalEp-Lawyer / u-para | 0.631 | 0.637 | 0.717 | 0.745 | 0.759 | 0.749 |
| LegalEp-DISC / u-para   | 0.551 | 0.551 | 0.618 | 0.648 | 0.648 | 0.642 |

Table A.5: CAIL2024 shared-corpus results ( $M \approx 3000$ ). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). \*\*: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$ . nDCG@10 uses gold-session IDCg.

| Channel     | System      | AH@10        | EC@10        | nDCG@10 |
|-------------|-------------|--------------|--------------|---------|
| U1          | FlatIP      | 0.803        | 0.544        | 0.538   |
|             | Soft O2     | 0.948        | 0.918        | 0.767   |
|             | Hard hydr.  | <b>0.995</b> | <b>0.994</b> | 0.859   |
|             | Shuffled O2 | 0.660        | 0.427        | 0.393   |
| Uk-followup | FlatIP      | 0.432        | 0.427        | 0.428   |
|             | Soft O2     | 0.932        | 0.925        | 0.796   |
|             | Hard hydr.  | <b>0.997</b> | <b>0.994</b> | 0.853   |
|             | Shuffled O2 | 0.313        | 0.348        | 0.349   |
| U-last      | FlatIP      | 0.393        | 0.415        | 0.400   |
|             | Soft O2     | 0.957        | 0.933        | 0.796   |
|             | Hard hydr.  | <b>0.998</b> | <b>0.995</b> | 0.861   |
|             | Shuffled O2 | 0.255        | 0.340        | 0.326   |

723        Soft O2 improves AH/EC over FlatIP; hard hydration uses unrestricted score copy  
724        and can lead both.

Table A.6: LegalEp-DISC ( $M=3000$ , 500 needles). Boldface: best AH@10 and best EC@10 within each channel. \*  $p < 0.05$ , \*\*  $p < 0.01$  (Soft O2 vs FlatIP, McNemar mid- $p$ ). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

| Channel       | System      | AH@10        | EC@10        | nDCG@10 |
|---------------|-------------|--------------|--------------|---------|
| exact         | FlatIP      | 0.714        | 0.857        | 0.700   |
|               | Soft O2     | 0.916        | 0.958        | 0.792   |
|               | Hard hydr.  | <b>1.000</b> | <b>1.000</b> | 0.873   |
|               | Shuffled O2 | 0.642        | 0.821        | 0.626   |
| advice-recall | FlatIP      | 0.472        | 0.560        | 0.454   |
|               | Soft O2     | 0.610        | 0.628        | 0.518   |
|               | Hard hydr.  | <b>0.648</b> | <b>0.648</b> | 0.563   |
|               | Shuffled O2 | 0.434        | 0.540        | 0.411   |
| u-param       | FlatIP      | 0.696        | 0.836        | 0.659   |
|               | Soft O2     | 0.883        | 0.932        | 0.742   |
|               | Hard hydr.  | <b>0.972</b> | <b>0.972</b> | 0.816   |
|               | Shuffled O2 | 0.620        | 0.797        | 0.590   |



Table A.7: LegalEp-Lawyer ( $M=3000$ , 500 needles). Boldface: best AH@10 and best EC@10 within each channel. \*  $p<0.05$ , \*\*  $p<0.01$  (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

| Channel       | System      | AH@10        | EC@10        | nDCG@10 |
|---------------|-------------|--------------|--------------|---------|
| exact         | FlatIP      | 0.678        | 0.839        | 0.710   |
|               | Soft O2     | 0.976        | 0.988        | 0.851   |
|               | Hard hydr.  | <b>1.000</b> | <b>1.000</b> | 0.880   |
|               | Shuffled O2 | 0.654        | 0.827        | 0.656   |
| advice-recall | FlatIP      | 0.416        | 0.546        | 0.457   |
|               | Soft O2     | 0.654        | 0.664        | 0.569   |
|               | Hard hydr.  | <b>0.674</b> | <b>0.674</b> | 0.591   |
|               | Shuffled O2 | 0.396        | 0.535        | 0.424   |
| u-param       | FlatIP      | 0.697        | 0.830        | 0.708   |
|               | Soft O2     | 0.956        | 0.965        | 0.832   |
|               | Hard hydr.  | <b>0.978</b> | <b>0.978</b> | 0.847   |
|               | Shuffled O2 | 0.675        | 0.815        | 0.656   |

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

| Corpus         | Channel                      | AH@10                 |
|----------------|------------------------------|-----------------------|
| LegalEp-DISC   | exact (BM25-turn)            | 0.802                 |
| LegalEp-DISC   | advice-recall (BM25-turn)    | 0.382                 |
| LegalEp-DISC   | exact (dense+BM25 RRF)       | 0.780                 |
| LegalEp-DISC   | advice (dense+BM25 RRF)      | 0.394                 |
| CAIL           | U1 / Uk / U-last (BM25-turn) | 0.748 / 0.482 / 0.413 |
| LegalEp-Lawyer | exact / advice (BM25-turn)   | 0.784 / 0.460         |